

Semantics: studying meanings and functions of words

Lexicology and Lexicography – **Course Website**

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Outline

- **Discussion of study questions** — Q1–Q7 on Hilpert, Saavedra, and Rains (2023)
- **Theoretical framework** — Quick review: Principle of No Synonymy, Distributional Hypothesis
- **COCA study overview** — Data, method, and results
- **Practice using Sketch Engine** — Analyse clipping pairs using collocations and word sketches

Preparatory reading: Hilpert,
Saavedra, and Rains (2023)

Study questions

Please read the paper and prepare answers to these questions for our class discussion:

1. What is the Principle of No Synonymy, and what does it predict about clippings and their source words?
2. How do the authors use the principle “You shall know a word by the company it keeps” (**Firth 1957**) to study meaning differences? What do they compare?
3. What does the paper find about clippings in “involved” vs. “informational” contexts? Which linguistic features distinguish them?
4. Some clippings (e.g., *fridge/refrigerator*) are nearly synonymous, while others (e.g., *cardio/cardiovascular*) show clear differences. What factors might explain this variation?
5. How can collocation analysis reveal semantic differences not obvious from intuition? Use an example from the paper.
6. Choose one pair from Table 1 (e.g., *exam/examination*). What differences might you expect, and what would you look for in a corpus analysis?
7. The paper offers two explanations for clippings in “involved” contexts: familiarity vs. efficiency. What evidence supports each? Which do you find more convincing?

Q1: Principle of No Synonymy

What is the Principle of No Synonymy, and what does it predict about clippings and their source words?

- **Definition:** A difference in linguistic form always indicates a difference in meaning.
- **Broad notion of “meaning”:** semantic content, but also discourse-functional, interpersonal, and information-structural aspects.
- **Prediction for clippings:** clippings and source words should have distinct distributional characteristics that reflect functional differences.

Example: *brother* (kinship term) vs *bro* (address term for close friend within specific communities) — clearly distinct meanings despite shared etymology.

Q2: Distributional Hypothesis

How do the authors use the principle “You shall know a word by the company it keeps” to study meaning differences? What do they compare?

Two levels of comparison:

1. **Aggregate level:** frequency of “involved” vs “informational” markers across all 50 pairs (Table 2)
2. **Individual level:** token-based semantic vector spaces using second-order collocates

Q3: Involved vs Informational Contexts

What does the paper find about clippings in “involved” vs “informational” contexts? Which linguistic features distinguish them?

The paper tests 12 markers of involved vs informational text production.

Involved markers (more frequent with clippings):

- Private verbs (*know, think, believe*)
- Contractions (*it's, you're*)
- Present tense verbs
- 1st and 2nd person pronouns (*I, you*)
- Indefinite pronouns (*something, anyone*)
- Pronoun *it*

Informational markers (more frequent with source words):

- Complementiser *that*
- Common nouns
- Prepositions

Q4: Variation in Synonymy

Some clippings (e.g., *fridge/refrigerator*) are nearly synonymous, while others (e.g., *cardio/cardiovascular*) show clear differences. What factors might explain this variation?

Clearly distinct pairs (Figure 3):

- *cardio/cardiovascular*: fitness vs medical contexts
- *chemo/chemotherapy*: patient experience vs technical medical
- *indie/independent*: underground culture vs general meaning
- *dorm/dormitory*: includes tourist lodging uses

Near-synonymous pairs (Figure 4):

- *fridge/refrigerator*: same semantic facets accessible to both
- *porn/pornography*: industry, psychological, legal aspects shared

Q5: Collocation Analysis

How can collocation analysis reveal semantic differences not obvious from intuition? Use an example from the paper.

Example: *cardio* vs *cardiovascular*

Intuition might suggest they mean the same thing. But collocates reveal:

***cardiovascular*:**

- *disease, diabetes, chronic*
- *renal, kidney, liver*
- Medical/clinical contexts

***cardio*:**

- *workout, fat-blasting, calendar*
- *recipes, oranges, dishes*
- Fitness/lifestyle contexts

Q6: Corpus Analysis of *exam/examination*

Choose one pair from Table 1 (e.g., *exam/examination*). What differences might you expect, and what would you look for in a corpus analysis?

Expected differences:

- *exam*: academic/medical test contexts, informal register
- *examination*: formal register, plus “investigation/scrutiny” sense

What to look for in a corpus:

- **Collocates**: *final exam, take an exam* vs *detailed examination, close examination, under examination*
- **Text types**: spoken/informal (blog, fiction) vs written/formal (academic, news)
- **Modifiers**: different adjectives? (*tough exam* vs *thorough examination*)
- **Syntactic patterns**: *examination of* (investigation) vs *exam in/on* (subject)?

Q7: Familiarity vs Efficiency

The paper offers two explanations for clippings in “involved” contexts: familiarity vs efficiency. What evidence supports each? Which do you find more convincing?

Familiarity explanation:

- Clippings signal common ground, shared expertise, in-group membership
- Evidence: co-occurrence with “involved” markers; personal experience contexts
- Example: *chemo* in patient narratives vs *chemotherapy* in medical papers

Efficiency explanation:

- Shorter forms preferred when predictable; saves processing effort
- Counter-evidence: predictability effects are limited when forms differ in meaning

Theoretical framework

Does clipping involve a change in meaning?

One view: clippings and source words are alternatives that only differ in form, not meaning.

Examples:

- *mic* vs. *microphone*
- *lab* vs. *laboratory*
- *ad* vs. *advertisement*

But: do *brother* and *bro* have the same meaning?

The Principle of No Synonymy

- A difference in linguistic form always indicates a difference in meaning (**Goldberg 1995**).
- “Meaning” is broad: semantic, discourse-functional, interpersonal, etc.
- **Prediction:** clippings and source words should have distinct distributions, reflecting functional differences.

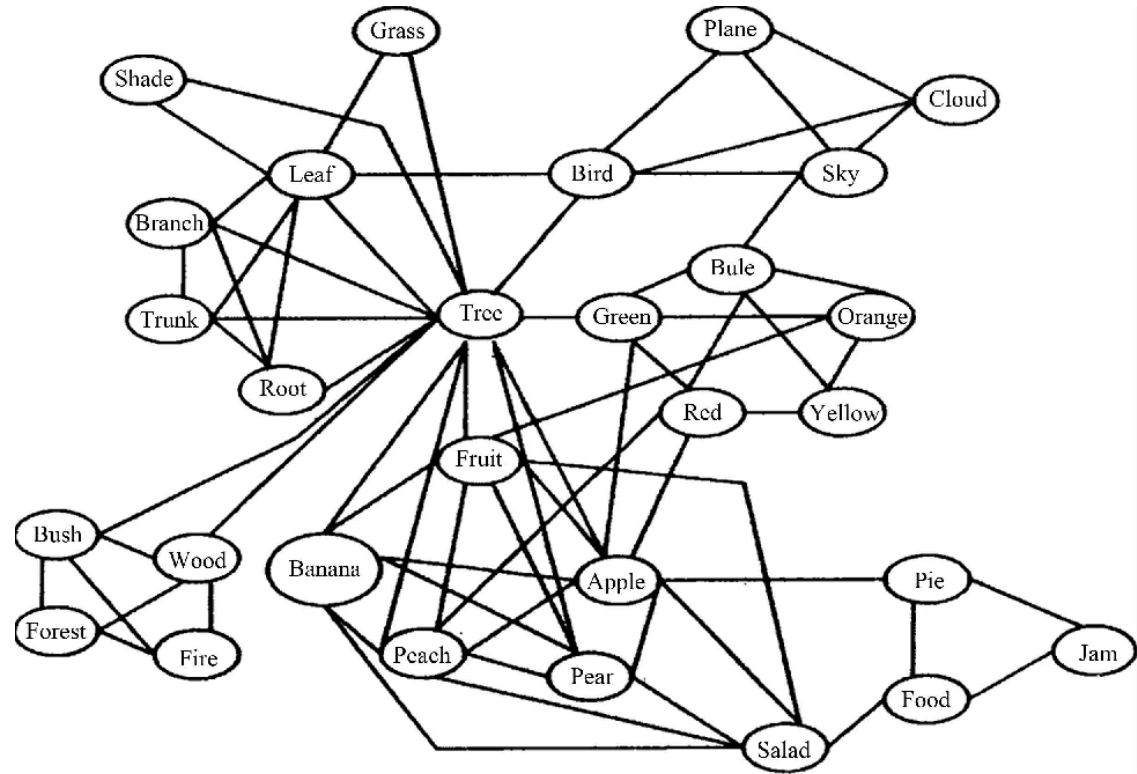
Note

Think of 3 sentences the words *brother* and *bro* could appear in.

The Distributional hypothesis

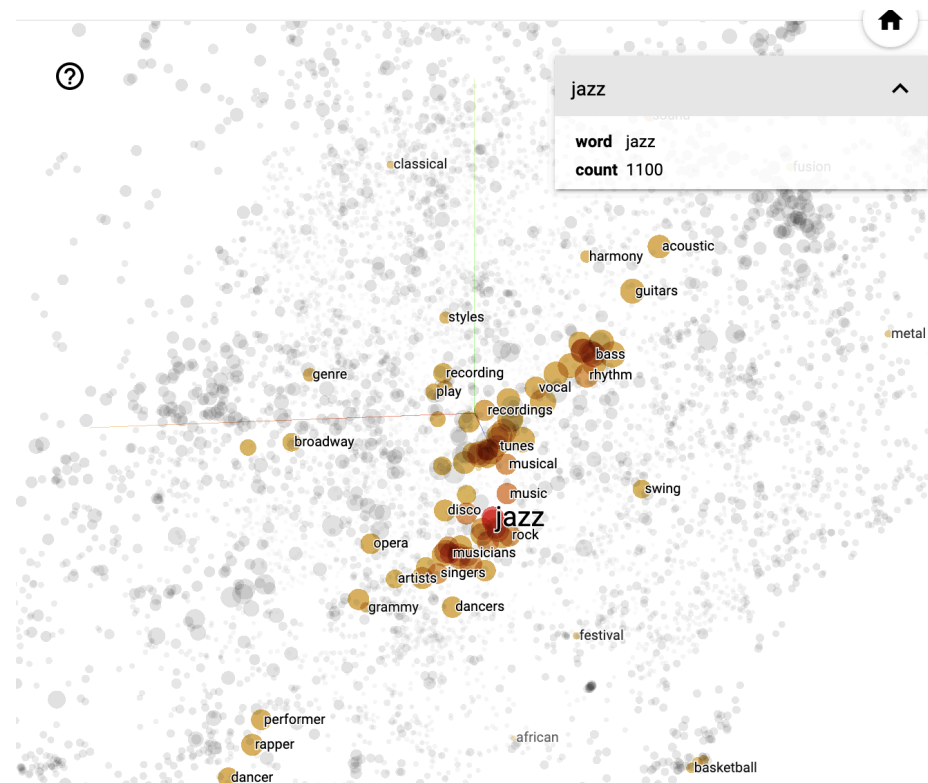
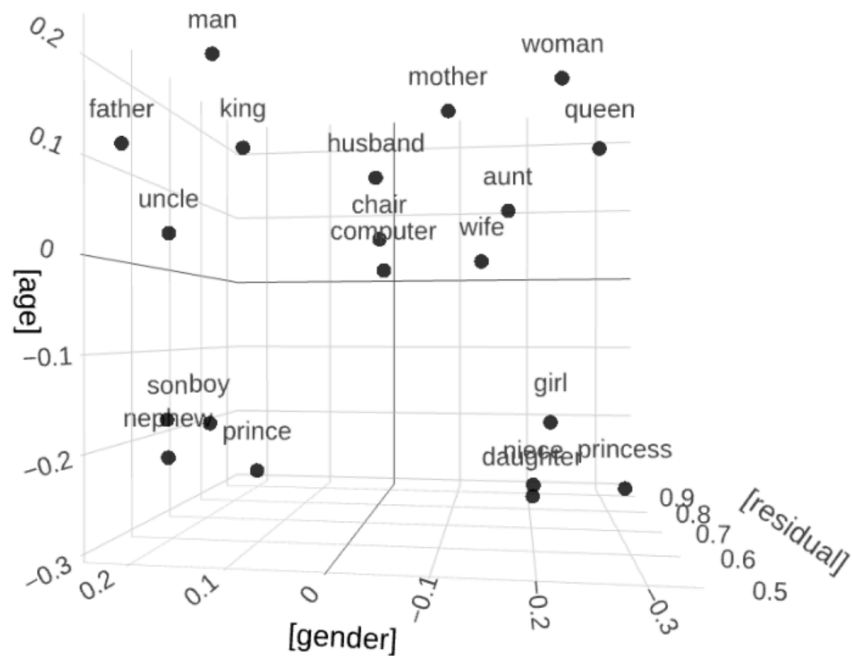
- The meaning of words is reflected in their contextual elements in language use: **“You shall know a word by the company it keeps”** (Firth 1957).
- Words that appear in similar contexts share aspects of their meanings.
- Example: *happy* and *joyful* share collocates like *very*, *feel*, *make*, *person*.

Cognitive-linguistic models of the lexicon



Word embeddings (Bandyopadhyay et al. 2022)

TensorFlow Projector



Exercise

Word Embeddings Explorer

1. Explore synonyms and antonyms (Similar words)

- Enter a seed word (e.g., **happy_ADJ**).
- Inspect the 5 nearest neighbours.
- Compare with an antonym (e.g., **sad_ADJ**).

2. Visualise semantic clusters (Visualization)

- Choose two related groups (e.g., animals vs furniture).
- Plot in vector space; describe visible groupings.

3. Word analogies (Calculator)

- Try analogies, e.g., **male_ADJ : doctor_NOUN = female_ADJ : ?**.
- Test several analogies; note successes and failures.

4. Investigate ambiguity

- Pick a polysemous word (e.g., *bank*).
- Analyse how neighbours reflect different senses.

Studying the meaning of clippings in the COCA

Hilpert, Saavedra, and Rains ([2023](#))

Data

- Sample of 50 English clippings and their corresponding source words.
- Corpus data from COCA ([Davies 2008](#)) for each clipping and source word.

Table 1. Selected clippings and their long forms.

	clipping	source word		clipping	source word		clipping	source word
1	admin	administration	18	expat	expatriate	35	mic	microphone
2	alum	alumnus	19	fab	fabulous	36	pic	picture
3	ammo	ammunition	20	fave	favorite	37	porn	pornography
4	amp	amplifier	21	frat	fraternity	38	prefab	prefabricated
5	app	application	22	fridge	refrigerator	39	prenup	prenuptial agreement
6	boomer	baby-boomer	23	gent	gentleman	40	prof	professor
7	burbs	suburbs	24	glam	glamorous	41	psych	psychology
8	carbs	carbohydrates	25	goth	gothic	42	pup	puppy
9	cardio	cardiovascular	26	hippo	hippopotamus	43	quake	earthquake
10	celeb	celebrity	27	improv	improvisation	44	rehab	rehabilitation
11	chemo	chemotherapy	28	indie	independent	45	roach	cockroach
12	chimp	chimpanzee	29	intro	introduction	46	stats	statistics
13	croc	crocodile	30	lab	laboratory	47	teen	teenager
14	decaf	decaffeinated	31	legit	legitimate	48	tux	tuxedo
15	dorm	dormitory	32	limo	limousine	49	undergrad	undergraduate
16	exam	examination	33	lube	lubricant	50	vid	video
17	exec	executive	34	memo	memorandum			

Method

Collocation analysis to investigate semantic profiles across contexts.

Table 4. Second-order collocates of tokens of *cardio* and *cardiovascular*.

Concordance line	Key word	Second-order collocates
(1a)	cardio	muscles, slam, abdominal, eyelids, stamina, circulation, nerves, headache, calves, ...
(1b)	cardio	recipes, oranges, cooked, pies, dishes, salads, sweets, foods, fried, dried, fruit, ...
(1c)	cardiovascular	diabetes, inflammatory, disorders, chronic, disease, renal, arthritis, kidney, liver, ...
(1d)	cardiovascular	chronic, obstruction, lung, asthma, epithelial, disease, hypertension, prevalence, intermittent, ...

Results

Variation across text types

Table 2. Twelve features of involved and informational text production (Biber 1988: 102).

Involved Text Production	Informational Text Production
private verbs*	complementizer THAT
contractions	common nouns
present tense verbs	prepositions
1st person pronouns	type/token ratio
2nd person pronouns	
indefinite pronouns**	
pronoun IT	
demonstrative pronouns	

*Following Biber (1988: 242), the verbs that were included in this category were *anticipate, assume, believe, conclude, decide, demonstrate, determine, discover, doubt, estimate, fear, feel, find, forget, guess, hear, hope, imagine, imply, indicate, infer, know, learn, mean, notice, prove, realize, recognize, remember, reveal, see, show, suppose, think, and understand*.

**Following Biber (1988: 226), the pronouns that were included in this category were *anybody, anyone, anything, everybody, everyone, everything, nobody, none, nothing, nowhere, somebody, someone, and something*.

Semantic differences

Differences in collocations for *cardiovascular* vs *cardio*

Table 4. Second-order collocates of tokens of *cardio* and *cardiovascular*.

Concordance line	Key word	Second-order collocates
(1a)	cardio	muscles, slam, abdominal, eyelids, stamina, circulation, nerves, headache, calves, ...
(1b)	cardio	recipes, oranges, cooked, pies, dishes, salads, sweets, foods, fried, dried, fruit, ...
(1c)	cardiovascular	diabetes, inflammatory, disorders, chronic, disease, renal, arthritis, kidney, liver, ...
(1d)	cardiovascular	chronic, obstruction, lung, asthma, epithelial, disease, hypertension, prevalence, intermittent, ...

Practice: studying meanings
using Sketch Engine

Your Task

1. Choose one of the word pairs from **the sample above** (Hilpert, Saavedra, and Rains 2023).
2. Use the collocations and word sketches in the **enTenTen21** corpus to analyse their meanings.
3. Use the **Guiding Questions** below to structure your analysis.
4. Take notes in **this PowerPoint file**.
5. Share your findings with the class.

Collocation analysis

1. Run a query

e.g. `[word="brother" & tag="N.*"]`

 SKETCH ENGINE

CQL builder

CQL: `[word="brother" & tag="N.*"]`

normal token

+

`[ word="brother" & tag="N.*" ]`

+



USE THIS CQL ➤

result example ▾

ly trio set for Belfast carol extravaganza </p><p> Eugene O'Hagan and his
ox (Joey) and Leigh Ann Williams (Kenneth). She was predeceased by her
y able. </p><p> He is survived by his parents Lisa and Chet Holowicki, his
Barista Ever, and his long-suffering manager Lestrade would fire him if his
ed time in prison with him for defying the apartheid government. "I've lost a
nch and husband, Ed (Peanut) and McAdoo L. Driver and wife Vickie; one
s sister told a school official that she had been abused and hadn't seen her
tlee Yoder (Carolyn) of Kalona, 11 grandchildren, 11 great grandchildren, a
oseph, helped him start the family-operated business at 4111 S. 6th St. His
<p> Promising Larkhill runner-up Token Bay, who is part-owned by Miller's

brother Martin grew up in Claudy, Co Londonderry. It was while they were students
brother : Bobby Smith. </p><p> Memorials may be made to Trenton United Metho
brother Drew Holowcki, and partner Joey Hinger. </p><p> Friends and family host
brother didn't happen to be the city health inspector... go wild. </p><p> Tears, sing
brother . My life is in a void, and I don't know who to turn to." </p><p> Talk show qu
brother , Adam Conseen of Va.; seven grandchildren; 23 great-grandchildren, 13 g
brother in months, the video said. </p><p> Jeremiah Oliver lived with his mother a
brother Hobert Yoder of Iowa City, and a sister Marjorie Waybill of Harrisonburg, Vi
brother , Ervin, later ran it with him and co-owned it, family members said. </p><p>
brother , Andrew, and Jill Miller's Sparkling Miss, a once raced five-year-old out of

2. Click the collocation analysis icon

CONCORDANCE English Web 2020 (enTenTen20)  [Get more space](#)     

CQL [word="brother"] • 2,433,001
56.42 per million tokens • 0.0056% 

            [KWIC](#)   

☐ Details Left context KWIC Right context

1	☐	 suntimes.com	at 38 on Saturday from a heart attack, his younger brother Matthew Morton revealed.	
2	☐	 societalsecurit...	vacy and the greater intrusion of new forms of 'big brother ' surveillance in people's daily lives.	
3	☐	 gilmerfreepress...	sister, was seen at a propaganda event beside her brother and his wife.	
4	☐	 dailymail.co.uk	aced as they reunited.	
5	☐	 wikidot.com	I expectant.	
6	☐	 wikidot.com	ress and they send you free stuff!"	
7	☐	 arkansasonline....	Monday that killed a college athlete and injured her brother . by Nyssa Kruse	
8	☐	 gilmerfreepress...	ers, Gladys Snyder and Clara Turner of Ohio; one brother , Junior McCumbers of Grafton, OH; 24 grandchild	

3. Configure window range and statistics

The screenshot shows the CONCORDANCE web application interface. At the top, the header includes the logo, the title "CONCORDANCE", a search bar containing "English Web 2020 (enTenTen20)", and various utility icons. Below the header, a status bar displays "CQL [word='brother'] • 2,433,001" and "56.42 per million tokens • 0.0056%". The main panel is titled "COLLOCATIONS" and has three tabs: "BASIC", "ADVANCED", and "ABOUT". The "BASIC" tab is active. It contains a search bar with "word" and a magnifying glass icon. To the right of the search bar is a "Range ?" section with a green box labeled "1" around it. This section includes a row of buttons: "-5", "-4", "-3", "-2", "-1", "KWIC", "1", "2", "3", "4", and "5". Below this row is a checkbox labeled "Custom range". To the right of the range section is a "Show functions ?" section with a green box labeled "2" around it. This section contains a list of checkboxes: "T-score" (checked), "MI" (checked), "MI3" (unchecked), "log likelihood" (unchecked), "min. sensitivity" (unchecked), "logDice" (checked), and "MI.log_f" (unchecked). At the bottom left, there is a "Minimum frequency in corpus ?" section with a value of "5" and a dropdown arrow. At the bottom right, there is a red circular button labeled "GO".

CONCORDANCE

English Web 2020 (enTenTen20)

CQL [word="brother"] • 2,433,001
56.42 per million tokens • 0.0056%

COLLOCATIONS

BASIC ADVANCED ABOUT

Attribute ?
word

Range ?

-5 -4 -3 -2 -1 KWIC 1 2 3 4 5

☐ Custom range

Show functions ?

- ☒ T-score
- ☒ MI
- ☐ MI3
- ☐ log likelihood
- ☐ min. sensitivity
- ☒ logDice
- ☐ MI.log_f

Minimum frequency in corpus ?
5

GO

4. View the results and test collocation measures



CONCORDANCE

CQL [word="brother"] • 2,433,001
56.42 per million tokens • 0.0056%

English Web 2020 (enTenTen20)

[Get more space](#) [Help](#) [Share](#) [Info](#)



KWIC

Collocations

[CHANGE CRITERIA](#) [BACK TO CONCORDANCE](#)

	Word	Cooccurrences ?	Candidates ?	T-score	MI	LogDice ↓	
1	☐ younger	142,791	1,474,707	377.66	10.75	10.23	...
2	☐ older	111,838	2,800,488	333.95	9.47	9.45	...
3	☐ sister	76,686	1,721,760	276.57	9.62	9.24	...
4	☐ elder	44,609	301,639	211.13	11.36	9.06	...
5	☐ twin	30,902	531,632	175.62	10.01	8.42	...
6	☐ his	695,214	99,786,308	827.04	6.95	7.80	...
7	☐ My	81,924	9,785,966	284.29	7.21	7.78	...
8	☐ His	67,572	9,307,687	257.93	7.01	7.56	...
9	☐ my	273,056	57,070,941	516.39	6.41	7.23	...

Word sketches: Single forms

1. Generate a word sketch for a clipped form (e.g. exam), specifying the word class.

WORD SKETCH

British National Corpus (BNC)



exam as noun 1,559x



CHANGE CRITERIA

BASIC

ADVANCED

AS A LIST

LEARN

Search ?

exam

Part of speech ?

auto

adjective

adverb

noun

verb

Subcorpus ?

none (the whole corpus)

Minimum frequency ?

auto

Minimum score ?

0



Translate ?

2. Examine syntactic contexts.

WORD SKETCH

English Web 2021 (enTenTen21)



[Get more space](#)

examination as noun 1,769,654x



modifiers of "examination"		
entrance	8.6	...
entrance examination		
thorough	8.5	...
a thorough examination of		
careful	8.3	...
careful examination of the		
cross	8.3	...
cross examination		
close	8.2	...
closer examination		
physical	7.9	...
physical examination		
microscopic	7.7	...
microscopic examination of		
forensic	7.6	...
forensic examination		

nouns modified by "examination"		
malpractice	7.1	...
examination malpractice		
finding	5.9	...
physical examination findings		
timetable	5.8	...
examination timetable		
glove	5.7	...
examination gloves		
hall	5.5	...
the examination hall		
fee	5.5	...
examination fee		
paper	5.4	...
examination papers		
syllabus	5.3	...
examination syllabus		

verbs with "examination" as object		
pass	7.9	...
passed the examination		
qualify	7.8	...
qualifying examination		
undergo	7.8	...
undergo a medical examination		
conduct	7.6	...
examination conducted		
administer	7.1	...
examination administered		
sit	6.5	...
sit the examination		
perform	6.5	...
examination was performed		
fail	6.2	...
failed the examination		

WORD SKETCH

English Web 2021 (enTenTen21)



Get more space +



exam as noun 2,131,195x



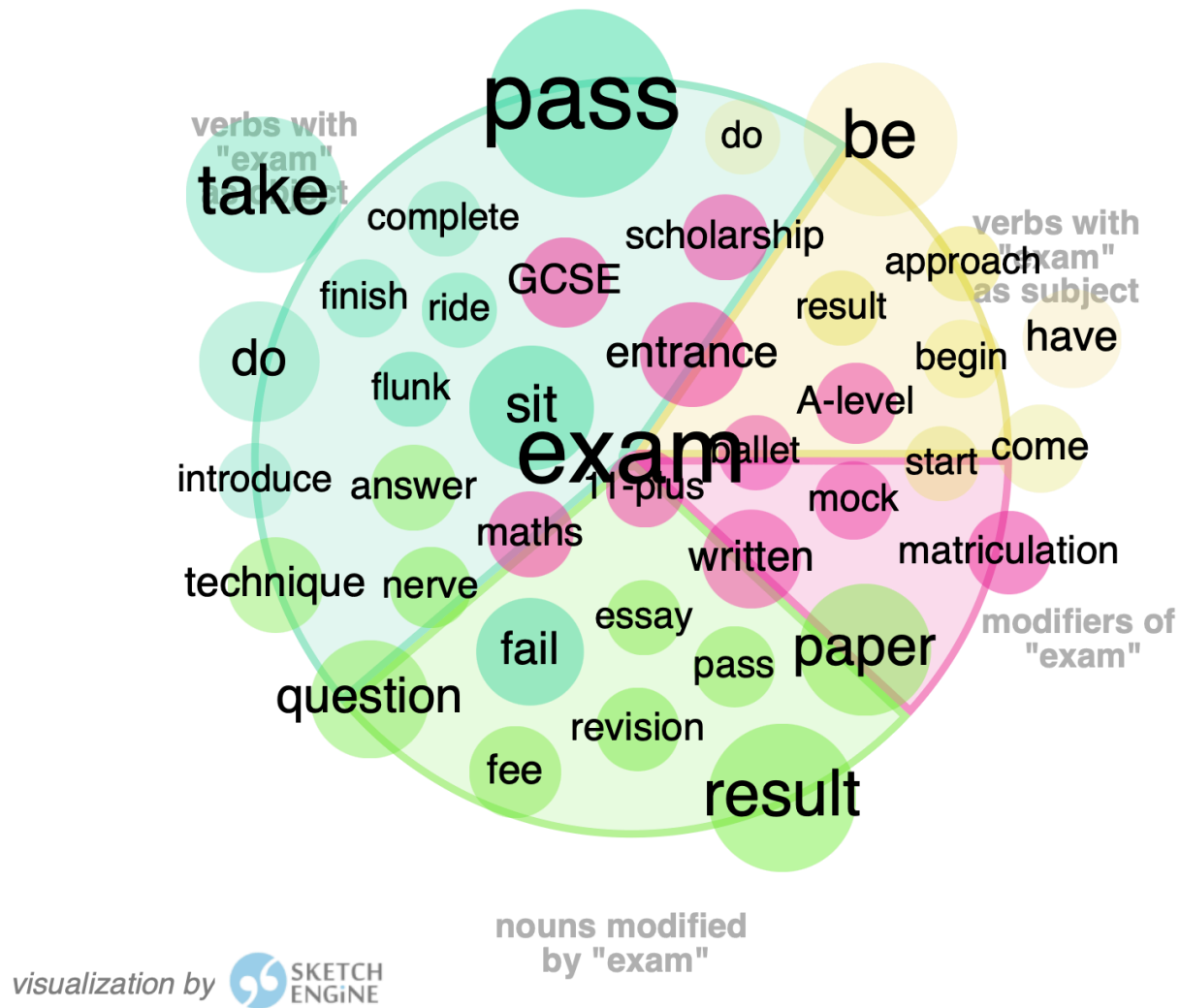
modifiers of "exam"		
certification	9.8	...
certification exam		
entrance	9.4	...
entrance exam		
practice	7.9	...
practice exams		
final	7.7	...
final exam		
eye	7.7	...
eye exam		
AP	7.6	...
AP exams		
bar	7.5	...
the bar exam		
written	7.5	...
written exam		

nouns modified by "exam"		
dump	9.0	...
exam dumps		
syllabus	8.4	...
exam syllabus . Pass4sure		
preparation	8.3	...
exam preparation		
braindump	8.2	...
exam braindump		
prep	8.1	...
exam prep		
question	7.5	...
exam questions		
Dumps	7.2	...
Real Certification Exam Dumps is Getting an		
engine	7.0	...
interactive exam engine		

verbs with "exam" as object		
pass	9.2	...
pass the exam		
sit	8.2	...
sit the exam		
fail	8.1	...
fail the exam		
practise	8.1	...
practise exam		
qualify	7.7	...
qualifying exam		
administer	7.5	...
exam administered		
retake	7.1	...
retake the exam		
schedule	7.0	...
exam is scheduled		

verbs with "exam" as subject		
dump	7.1	...
exam dumps		
file	6.3	...
can get all exam files paying an astonishingly		
test	5.8	...
exam tests		
assess	5.7	...
exam assesses		
consist	5.4	...
exam consists of		
require	5.2	...
Exam requires the candidates to		
loom	5.2	...
exams looming		
grade	5.2	...
exam grading		

3. Visualise results.



Word sketches: Comparison

1. Compare source words and clipped forms (e.g. examination vs exam).

 CHANGE CRITERIA

BASIC

ADVANCED

LEARN 

compare ? 

☒ Lemmas

☐ Word forms

☐ Subcorpora

First lemma ?

Second lemma ?

examination

exam

Part of speech ?

auto

adjective

adverb

noun

verb

2. Review shared and unique collocates.

examination 6,046x

exam 1,559x

adjective predicates of "examination/exam"

... is a "examination/exam"

verbs with "examination/exam" as subject

↔

⚙️

🔍

✕

"examination/exam" and/or ...

assessment	42	0	...
examination	20	0	...
curriculum	16	0	...
dissertation	7	0	...
biopsy	7	0	...
course	12	3	...
test	26	8	...
essay	5	3	...
er	0	3	...
work	0	3	...
something	0	4	...
exam	0	8	...

↔

⚙️

🔍

✕

verbs with "examination/exam" as object

undergo	28	0	...
resit	9	0	...
conduct	28	0	...
perform	23	0	...
finish	6	4	...
take	89	74	...
sit	43	35	...
do	23	31	...
fail	20	20	...
pass	83	128	...
ride	0	3	...
flunk	0	3	...

↔

⚙️

🔍

✕

modifiers of "examination/exam"

post-mortem	60	0	...
histological	58	0	...
detailed	130	0	...
close	147	0	...
mortem	35	0	...
medical	125	4	...
entrance	35	18	...
matriculation	9	6	...
GCSE	10	9	...
scholarship	6	7	...
11-plus	0	4	...
maths	0	7	...

3. Inspect collocations in detail.

CONCORDANCE

British National Corpus (BNC)



Get more space +



CQL **er + exam** • 3

0.03 per million tokens • 0.0000027%



KWIC ▾



☐ Details

Left context

KWIC

Right context

- | | | | | |
|---|--------------------------|--|---|--|
| 1 | ☐ | | Spoken context-... ow is that you will need it for your er , your Q M exam not er your exam, your project right, cos there is | |
| 2 | ☐ | | Spoken context-... ould I like to do it and so of course I had to pass exams and er actually, can I read some notes that | |
| 3 | ☐ | | Spoken context-... all of course er I kn I know about take two more exams and er I know the second one was mental arithn | |

Guiding questions

1. What is the general semantic signature of the source word?
2. What is the general semantic signature of the clipped form?
3. In what ways do they differ (stylistic, social, formality)?
4. Do they occur in different syntagmatic contexts despite similar denotation?
5. Does the clipped form have a wider or narrower scope of meaning?

Summary

- **A difference in form (often) implies a difference in meaning.** (Principle of No Synonymy)
- **Meaning is use.** The contexts a word appears in tell us about its function. (Distributional Hypothesis)
- **Clippings are not just lazy shortcuts.** They develop unique semantic, pragmatic, and social functions, often signalling familiarity and appearing in more “involved” contexts.

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